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Two important thms.

Thm (Euler) Let m be a positive integer and let

$a \in \mathbb{Z}$ with $(a, m) = 1$. Then

$$a^{\phi(m)} \equiv 1 \pmod{m}.$$

Thm (Fermat) Let p be prime and $a \in \mathbb{Z}$ with $p \nmid a$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, $a^p \equiv a \pmod{p}$.

E.g. Simplify $2^{100} \pmod{13}$.

$p=13$ is prime, $100 = 12 \cdot 8 + 4$.

$$\text{Fermat} \Rightarrow 2^{100} = (2^{12})^8 \cdot 2^4 \equiv 1^8 \cdot 16 \equiv 3 \pmod{13}.$$

Solving linear congruence equations

Thm. The linear equation $ax \equiv b \pmod{m}$ has

a solution if and only if $d \mid b$, where $d = \gcd(a, m)$,

in which case there are exactly d solutions. Moreover,

if x_0 is a solution, then every solution is of the form

$x = x_0 + k \cdot \frac{m}{d}$, where k is any integer.

E.g. Solve $12x \equiv 18 \pmod{42}$.

$d = (12, 42) = 6$, $d \mid 18$. So we have 6 solutions.

Dividing by 6: $2x \equiv 3 \pmod{\frac{42}{6}}$. $x_0 = 5$ is a solution.

So all the solutions are $x \equiv 5, 12, 19, 26, 33, 40 \pmod{42}$.

More generally, one may study polynomial congruences

$$f(x) \equiv 0 \pmod{m},$$

where $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ with coefficients

$a_0, a_1, \dots, a_n \in \mathbb{Z}$. (The set $\mathbb{Z}[x]$ of all polynomials with

integer coefficients is a commutative ring under $+$ and $\times$,

and in fact, a UFD)

Thm (Lagrange) Let p be a prime and

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \in \mathbb{Z}[x]$$

with $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has

at most n roots.

Pf. Exercise. Hint: Induction on n .

Note: Lagrange's thm may fail for a composite modulus:

$x^2 - 1 \equiv 0 \pmod{8}$ has 4 roots: $x \equiv 1, 3, 5, 7 \pmod{8}$.

Quadratic congruence equations.

A quadratic congruence equation in unknown x is of the form

$$ax^2 + bx + c \equiv 0 \pmod{m},$$

where $a \not\equiv 0 \pmod{m}$.

Q: How to solve it? Think of

$$ax^2 + bx + c = 0.$$

Completing the square:

$$a(x^2 + \frac{b}{a}x) + c = 0$$

$$a \left(x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 \right) + c = 0$$

$\underbrace{\hspace{10em}}$
 $\left(x + \frac{b}{2a}\right)^2$

$$a \left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a} - c = \frac{b^2 - 4ac}{4a}$$

Observations: 1. Division is tricky and makes coefficients

no longer integers.

2. Don't know if we can take $\sqrt{\quad}$.

Reduction: Multiplying $ax^2 + bx + c \equiv 0 \pmod{m}$ by $4a$:

$$\underline{4a^2x^2 + 4abx} + 4ac \equiv 0 \pmod{4am}.$$

$$\underbrace{4a^2x^2 + 4abx + b^2}_{(2ax+b)^2} - b^2 + 4ac \equiv 0 \pmod{4am}$$

$$(2ax+b)^2$$

Δ

$$(2ax+b)^2 \equiv \overbrace{b^2 - 4ac}^{\Delta} \pmod{4am}$$

Let $y = 2ax + b$. Then this is

$$y^2 \equiv \Delta \pmod{4am}.$$

Our task reduces to solving something like.

$$x^2 \equiv a \pmod{m}.$$

Q: Solvable? If yes, how many solutions?

We will focus on $m = p$ is prime.

Lem. Let $p > 2$ be prime and $a \in \mathbb{Z}$ s.t. $p \nmid a$. Then

$$x^2 \equiv a \pmod{p}$$

has no solution or 2 solutions.

Pf. Assume it has a solution x_0 . Let x be any solution. Then

$$x_0^2 \equiv a \pmod{p} \quad \textcircled{1}$$

$$x^2 \equiv a \pmod{p}. \quad \textcircled{2}$$

$$\textcircled{2} - \textcircled{1}: \quad x^2 - x_0^2 \equiv 0 \pmod{p}.$$

$$\Rightarrow (x + x_0)(x - x_0) \equiv 0 \pmod{p}.$$

S. Euclid $\Rightarrow x + x_0 \equiv 0 \pmod{p}$ or $x - x_0 \equiv 0 \pmod{p}$.

So $x \equiv -x_0 \pmod{p}$ or $x \equiv x_0 \pmod{p}$.

Note that $-x_0 \not\equiv x_0 \pmod{p}$.

Why: If $-x_0 \equiv x_0 \pmod{p}$, then $0 \equiv 2x_0 \pmod{p}$

Euclid $\Rightarrow p/2$ or p/x_0 , impossible because $p > 2$,

and $p \nmid a$ and $x_0^2 \equiv a \pmod{p}$. □

Goal: Give a prime $p > 2$, determine all the values of

$a \in \mathbb{Z} \ (p \nmid a)$ for which $x^2 \equiv a \pmod{p}$ is solvable.

Defn (Quadratic residues & nonresidues). An integer a , indivisible

by p , is a quadratic residue mod p if there is

$x \in \mathbb{Z}$ s.t. $x^2 \equiv a \pmod{p}$. Otherwise, a is called a quadratic nonresidue mod p .

Notation: Write $a \in R_p$ if a is a residue mod p ,
 $a \in N_p$ if a is a nonresidue mod p .

E.g. $1 \in R_3$: $x=1$ solves $x^2 \equiv 1 \pmod{3}$.

$-1 \in N_3$: $x^2 \equiv -1 \pmod{3}$ is unsolvable:

$$1^2 \equiv 1 \pmod{3}, \quad 2^2 \equiv 1 \pmod{3}.$$